

3.3 CAPACITATED OPTIMAL FLOW TRANSPORT WITH ENTROPIC REGULARIZATION

To align with the minimum cost flow problems, in this subsection, we consider adding constraints to our OFT so that our algorithm can address the minimum cost flow problem with matrix-vector iterations. The detailed algorithm is shown in Alg. 2 in Appendix B. Next, we will categorize the capacity constraints into node constraints and edge constraints, and subsequently refine our OFT-Sinkhorn algorithm accordingly.

Capacitated Constraints on Nodes Initially, we consider Capacitated constraints on nodes, which involve imposing constraints on the optimization in Eq. 5: $\mathbf{q} \leq \mathbf{r}$ and $\mathbf{q} - \mathbf{s} \leq \mathbf{r}$, where $\mathbf{r}$ represents the maximum capacity for incoming or outgoing flows at nodes. To address these capacity constraints in the OFT-Sinkhorn Algorithm modification, we perform a truncation operation, specifically setting $\mathbf{q}^{l+1} \leftarrow \min(\mathbf{r}, \min(\mathbf{r} - \mathbf{s}, \mathbf{q}^{l+1}))$, where $\mathbf{q}^{l+1}$ is defined in Eq. 9 for the $(l+1)$ -th iteration within the OFT-Sinkhorn Algorithm.

Capacitated Constraints on Edges Then we examine the capacitated constraints on edges, which replace the constraints $\text{diag}(\mathbf{P}) = \mathbf{d}$ in Eq. 5 with the constraints $\mathbf{P} \leq \mathbf{S}$, where $\mathbf{S}_{ii} = d$ represents the constraints of virtual flow and $\mathbf{S}_{ij}$ denotes the capacity for edge e_{ij} . By employing the Lagrangian method, we can derive the following proposition, with the proof provided in the Appendix C.2.

Proposition 3. *The solution of entropic OFT with edge capacity is unique and has the form:*

$$\mathbf{P}^* = \text{Diag}(\mathbf{u})\mathbf{K}\text{Diag}(\mathbf{v}) \quad \text{and} \quad \mathbf{q}^* = \frac{\mathbf{s}}{2} + \left((\mathbf{K}\mathbf{v}) \odot (\mathbf{K}^\top \mathbf{u}) + \frac{\mathbf{s}^2}{4} \right)^{\frac{1}{2}} - \mathbf{d} \quad (13)$$

Here $\mathbf{u}, \mathbf{v}$ are two (unknown) scaling variables satisfying $\mathbf{u} \odot \mathbf{v} = \mathbf{1}$ and $\mathbf{K} = e^{(-\mathbf{D}+\mathbf{G})/\epsilon}$ where $\mathbf{G}$ is defined as

$$\mathbf{G} = \min(\mathbf{D} + \epsilon \cdot \log(\text{Diag}(\mathbf{u}^{-1})\mathbf{S}\text{Diag}(\mathbf{v}^{-1})), \mathbf{0}) \quad (14)$$

For the Capacitated Edges Constraints in the OFT-Sinkhorn algorithm, we will modify the iteration for $\mathbf{K}$ in Eq. 10 as follows:

$$\mathbf{K}^{(l+1)} = \min(\mathbf{K}, \text{Diag}(\mathbf{u}^{(l+1)})^{-1}\mathbf{S}\text{diag}(\mathbf{v}^{(l+1)})^{-1}) \quad (15)$$

By iteratively applying Eq. 8, Eq. 9, and Eq. 15 until convergence, we can derive the modified OFT-Sinkhorn algorithm based on edge capacity constraints.

Capacitated constraints on nodes and edges are typically meaningful in practical scenarios. For instance, in transportation problems where edges represent roads and nodes denote intersections at either end of the roads, node capacity constraints can effectively express the limit of vehicles passing through intersections. Meanwhile, edge capacity constraints can depict the maximum flow allowed on each road, thereby enabling our capacitated OFT to more accurately model transportation issues within a traffic road network.

4 EXPERIMENT

4.1 DATASETS AND EXPERIMENT SETTINGS

We conduct experiments using an NVIDIA GeForce RTX 4090 GPU, with programs implemented in Python and PyTorch. We evaluate our approach on three graph datasets: Uniform-MCF, NETGEN-MCF and Vision-MCF. For all experiments except the ablation study, we use the same settings to demonstrate our method’s robustness. Since only our matrix iteration algorithm can run efficiently on GPU, the other three comparison methods were tested on the Intel i9-10920X CPU.

Datasets We adopt the following datasets in our experiments:

- **Uniform-MCF (Double-precision):** We consider the symmetry between source and target nodes(i.e., 10% of nodes designated as sources and targets), constructing four synthetic datasets accordingly(w/o constraint, w/edge constraint, w/node constraint, w/edge+node constraint). We utilizes Gaussian function to generate double-precision point within Euclidean space. The initial flow of source and target nodes are generated by Uniform function which is also double-precision. The edge capacity are set to the same value on the same set of parameters. For more details, please refer to Appendix D.